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548113 (37)

ME/M Tech. (1st Semester)

Examination, Nov-Dec., 2014

Branch : Mechanical Engg.

Specialization : Design

ADVANCED DYNAMICS OF MACHINE

Time Allowed : Three Hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note : (1) Attempt any two parts in each question.

(2) All questions carry equal marks.

Q. 1. (a) Discuss the analysis of four-link mechanism with an example. **10**

(b) The dimensions of a four - link mechanism are AB = 500 mm, BC = 660 mm, CD = 560

mm and AD = 1000 mm. The link AB has an angular velocity of 10.5 rad/s counter-clockwise and an angular retardation of 26 rad/s^2 at the instant when it makes an angle of 60^0 with AD, the fixed link. The mass of the links BC and CD is 4.2 kg/m length. The link AB has a mass of 3.54 kg, the centre of which lies at 200 mm from A and a moment of inertia of 88500 kg-mm^2. Neglecting gravity and friction effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces. **10**

(c) Derive a relation for the turning moment at the crankshaft in terms of piston effort and the angle turned by the crank. **10**

Q. 2. (a) What do you mean by dynamically equivalent system? In what way the inertia of the connecting rod of a reciprocating engine is taken into account? **10**

(b) The following data relate to a horizontal reciprocating engine :

Mass of reciprocating parts = 120 kg,

Crank length = 90 mm, engine speed = 600 rpm

For connecting rod :

Mass = 90 kg

Length between centres = 450 mm

Distance of centre of mass from big end centre = 180 mm.

Radius of gyration about an axis through centre of mass = 150 mm.

Find the magnitude and the direction of the inertia torque on the crankshaft when the crank has turned 30° from the inner - dead centre. **10**

(c) Describe the graphical method of considering the inertia of the connecting rod of a reciprocating engine. **10**

Q. 3. (a) Explain the undamped response in dynamic force analysis of the force - closed cam follower. **10**

(b) Explain the analysis of elastic cam systems. **10**

(c) Describe unbalance, spring surge and windup. **10**

Q. 4. (a) Derive the expression for single rotor **10**

$$y = \frac{(w / w_c)^2 e}{1-(w / w_c)^2}$$ where

y = deflection of geometric centre from axis of rotation, due to centrifugal force.

w = angular speed of the shaft

w_c = critical speed of shaft.

e = eccentricity of the rotor.

(b) A rotor having a mass of 6 kg is mounted midway on a simply supported shaft of

diameter 10 mm and length 400 mm. The centre of gravity of rotor is 0.02 mm away from the geometric center of the rotor. If the rotor rotates at 2500 rpm, find the amplitude of steady state vibrations and the dynamic force transmitted to the bearings. Assume for shaft material, E = 200 GPa. **10**

(c) Find the expression for natural frequency for the critical speed of shaft carrying multiple rotors using Rayleigh's method. **10**

Q. 5. (a) Explain the first order system response for a linear system. **10**

(b) A second order measurement system has an effective mass of 1.2 kg, the stiffness

coefficient 60 kN/m and the damping factor 0.42. Assuming a unit step input, find : **10**

(i) the rise time.

(ii) the setting time for a tolerance band of 2%

(c) What are the open and closed loop control systems? Explain by giving examples. **10**